Class 5

Reverse Proxy and Integration

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Flow

	Frontend
Install package	Install **nginx** from nginx.org
[Start service]	`$ systemctl start nginx`
Install source	`$ tar` `$ rsync`
Configure the service	`$ micro config.js` `$ micro /etc/nginx/conf.d/site.conf`
Run the service	`$ nginx -t` `$ systemctl reload nginx`

Flow

1. Install package
2. [Start service]
3. Install source
4. Configure the service
5. Run the service

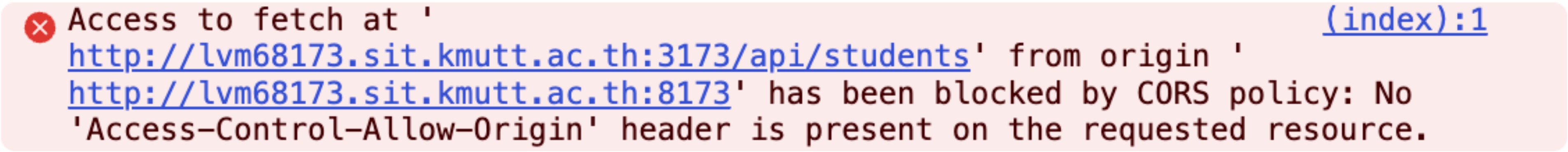
Database

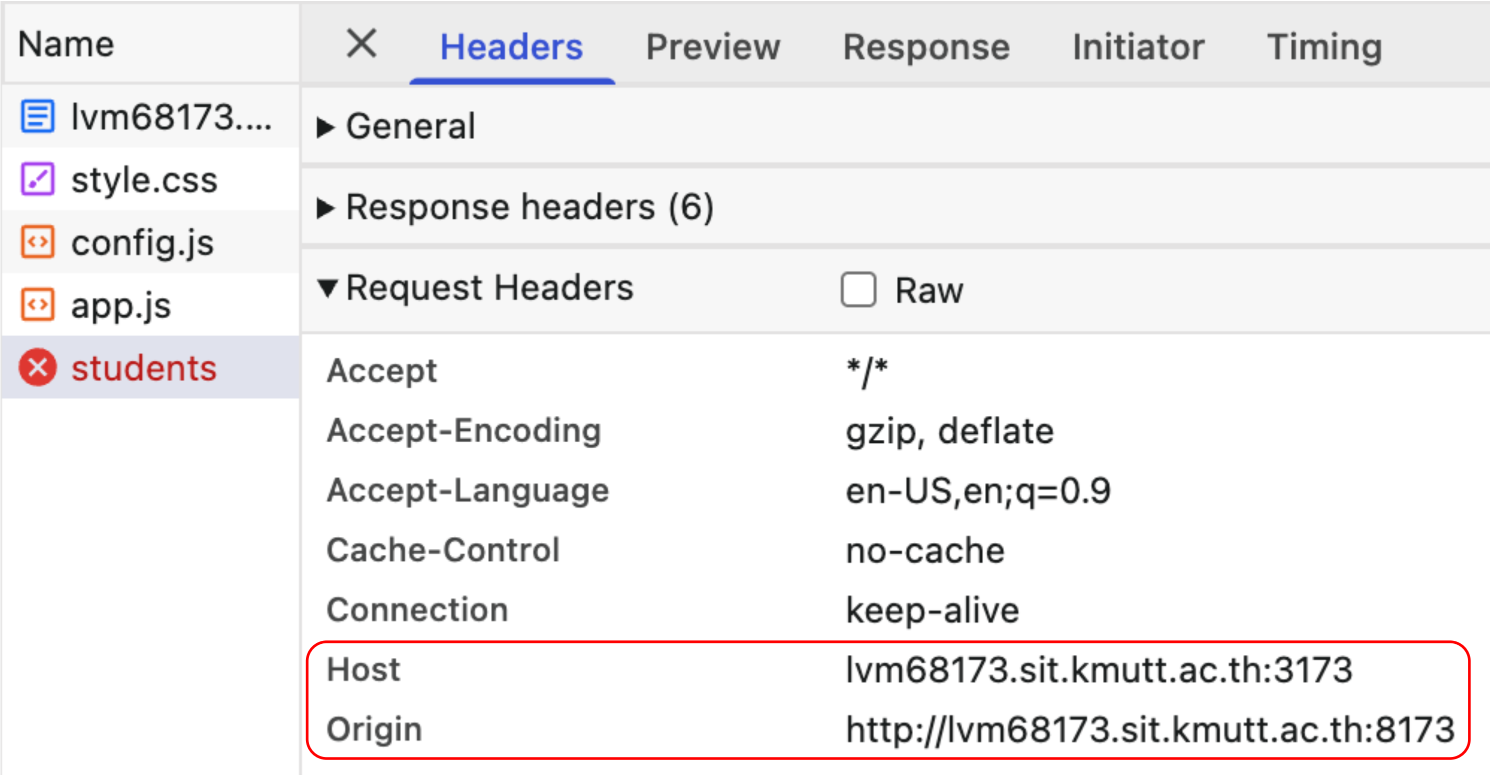
1. Install **`mysql-server`**
2. `$ systemctl status mysql`
3.
4. ```
   $ sudo mysql < setup.sql
   $ ... < schema.sql
   $ ... < seed.sql
   ```
5.

Backend

1. Install **`nodejs`** from nodesource
2.
3. `$ npm ci`
4. ```
   $ micro .env
   $ npx prisma generate
   ```
5. ```
   $ micro /etc/systemd/system/api.service
   $ systemctl enable --now api
   $ ufw allow <port>/tcp
   ```
   ```

Problem with CORS





Cross-Origin Resource Sharing (CORS)

“ Cross-Origin Resource Sharing (CORS) is an HTTP-header based mechanism that allows a server to indicate any origins (domain, scheme, or port) other than its own from which a browser should permit loading resources. ”

https://developer.mozilla.org/en-US/docs/Web/HTTP/CORS

“ CORS defines a way in which a browser and server can interact to determine whether it is safe to allow the cross-origin request. It allows for more freedom and functionality than purely same-origin requests, but is more secure than simply allowing all cross-origin requests. ”

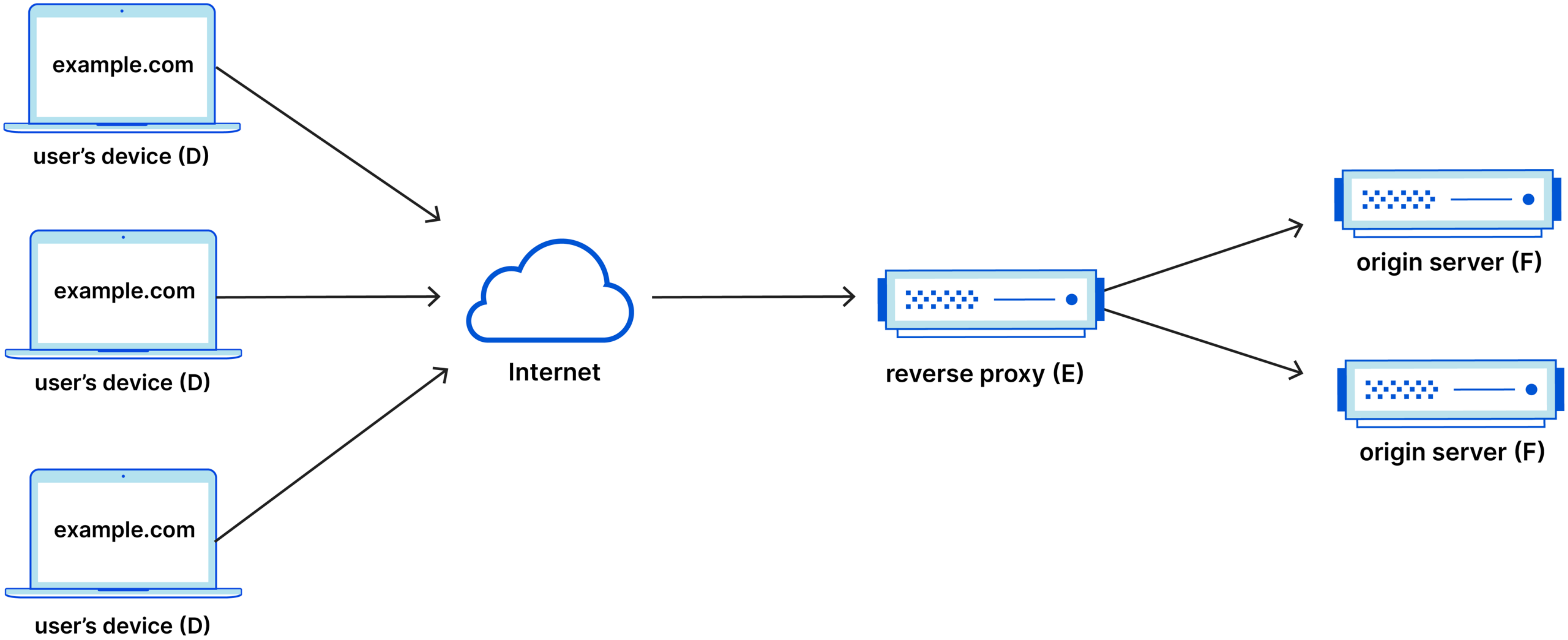
https://en.wikipedia.org/wiki/Cross-origin_resource_sharing

Reverse Proxy

- A reverse proxy is a server that (transparently) forwards the client's requests to one or more web servers
- A reverse proxy sits in front of an origin server and ensures that no client ever communicates directly with that origin server
- Reverse proxies help increase scalability, performance, resilience, and security

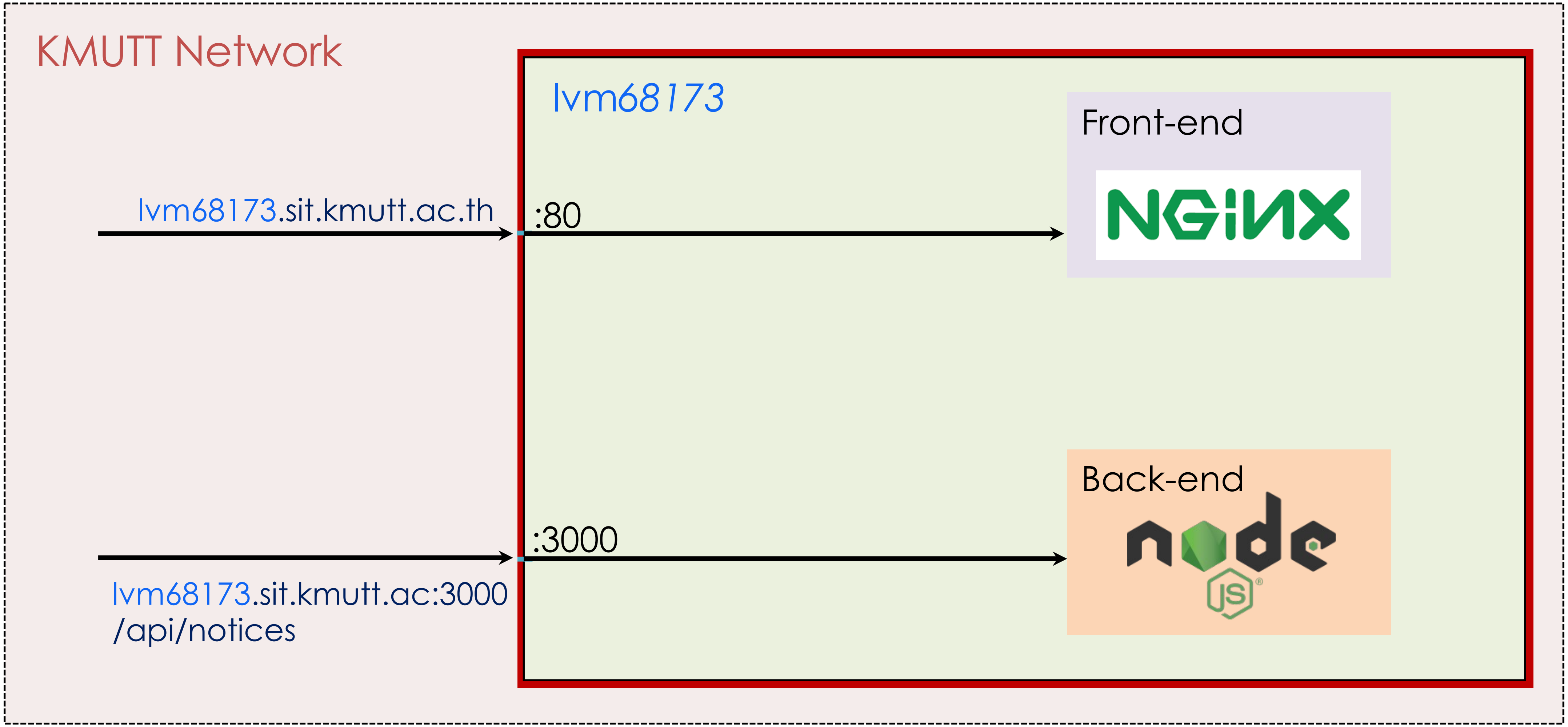
https://en.wikipedia.org/wiki/Reverse_proxy
https://www.cloudflare.com/en-gb/learning/cdn/glossary/reverse-proxy/

Reverse Proxy

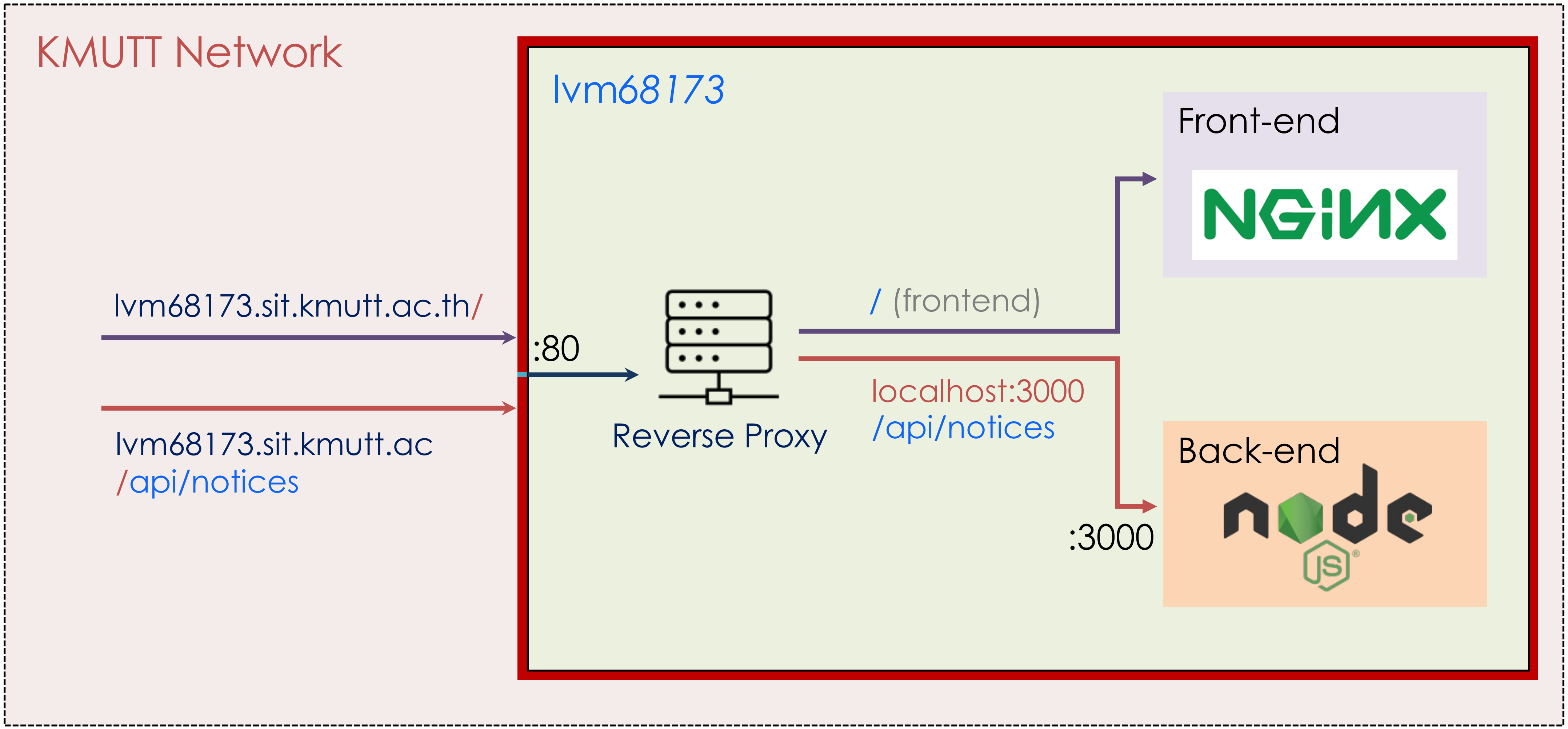


https://www.cloudflare.com/en-gb/learning/cdn/glossary/reverse-proxy/

Without Reverse Proxy



With Reverse Proxy



With Reverse Proxy and Nested Path